Vector Integration

❖ Ordinary integrals of vectors

Let $\mathbf{R(u)} = R_1(u)\,i + R_2(u)\,j + R_3(u)k$ be a vector depending on a single scalar variable *u*, the *indefinite integral* of **R(u)** is given:

$$\int \mathbf{R(u)}du = i\int R_1(u)du + j\int R_2(u)du + k\int R_3(u)du$$

If there exists a vector **S(u)** such that $\mathbf{R(u)} = \frac{d}{du}(\mathbf{S(u)})$, then

$$\int \mathbf{R(u)}du = \int \frac{d}{du}(\mathbf{S(u)})du = \mathbf{S(u)} + c$$

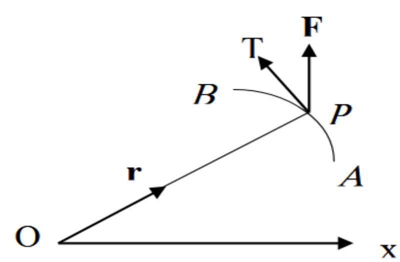
where *c* is an arbitrary constant vector independent of *u*.

The *definite integral* between limits *u=a* and *u=b* can be written

$$\int_a^b \mathbf{R(u)}du = \int_a^b \frac{d}{du}(\mathbf{S(u)})du = \mathbf{S(u)} + c|_a^b = S(b) - S(a)$$

❖ Line integrals

Let $\mathbf{F}(x,y,z) = F_1 i + F_2 j + F_3 j$ be a vector function and a curve *AB*. Line integral of a vector function **F** along the curve *AB* is defined as integral of the component of **F** along the tangent to the curve *AB*.



Component of **F** along a tangent *PT* at *P* equal (Dot product of **F** and unit vector along *PT*) $= \mathbf{F} \cdot \frac{d\mathbf{r}}{ds}$

Line integral $= \sum \mathbf{F} \cdot \frac{dr}{ds}$ from A to B along the curve.

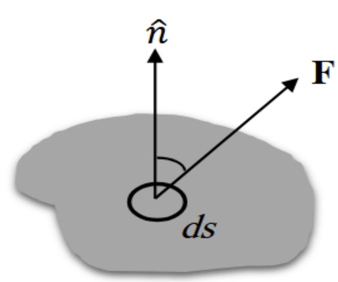
$$\therefore Line\ integral = \int_c \left(\mathbf{F} \cdot \frac{d\mathbf{r}}{ds}\right) ds = \int_c \mathbf{F} \cdot d\mathbf{r}$$

Notes:

1. When the path of integration is a closed curve then notation of integration is $\oint$ in place of $\int$. (***Closed curve****: a curve which does not intersect itself anywhere*.)
2. If $\mathbf{F}$ represents the variable force acting on a particle along arc AB, then the total *work* done $= \int_A^B \mathbf{F} \cdot \mathrm{d}\mathbf{r}$.
3. If $\mathbf{v}$ represents the velocity of a liquid then $\oint_c \mathbf{v} \cdot d\mathbf{r}$ is called the *circulation* of $\mathbf{v}$ round the closed curve c.
4. A vector field $\mathbf{F}$ is conservative if and only if $\nabla \times \mathbf{F} = 0$, or equivalently $\mathbf{F} = \nabla\Phi$. In such case $\mathbf{F}$ is called a *conservative vector field* and Φ is its *scalar potential*. Hint: $[\mathrm{d}\Phi = \nabla\Phi \,.\, dr = \mathbf{F} \cdot \mathrm{d}\mathbf{r}]$

❖ Surface integrals

Let $\mathbf{F}(x, y, z) = F_1 i + F_2 j + F_3 j$ be a vector function and S be the given surface. Surface integral of a vector function $\mathbf{F}$ over the surface S is defined as integral of the component of $\mathbf{F}$ along the normal to the surface.



Component of $\mathbf{F}$ along the normal $= \mathbf{F} \cdot \hat{n}$, where $\hat{n}$ is the unit normal vector to an element ds and

$$\hat{n} = \frac{\nabla f}{|\nabla f|} \quad , \qquad ds = \frac{dx\ dy}{(\hat{n} \,.\, \hat{k}\,)}$$

Surface integral of $\mathbf{F}$ over S

$$= \sum \mathbf{F} \cdot \hat{n} \quad = \iint_S (\mathbf{F} \cdot \hat{n})\ ds$$

Notes:

1. The notation $\oiint$ or $\oint_S$ is used to indicate integration over the closed surface S.
2. A unit normal $\hat{n}$ to any point of the positive side of S is called a *positive* or *outward drawn* unit normal.
3. Flux $= \iint_S (\mathbf{F} \cdot \hat{n})\, ds$ where, **F** represents the velocity of a liquid.
4. If $\iint_S (\mathbf{F} \cdot \hat{n})\, ds = 0$, then **F** is said to be a *solenoidal* vector point function.

❖ **Volume integrals**

Let $\mathbf{F}(x, y, z) = F_1 i + F_2 j + F_3 j$ be a vector function and volume V enclosed by a closed surface.

The volume integral of **F** over V

$$= \iiint_S \mathbf{F}\, dV$$

Examples:

Ex. (1): Find the total work done in moving a particle in a force field given by $\mathbf{F} = 3xy\, i - 5z\, j + 10x\, k$ along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from t = 1 to t = 2 ?

Solution: total *work* done = $\int_C \mathbf{F} \cdot \mathbf{dr}$

$$= \int_C (3xy\, i - 5z\, j + 10x\, k) \cdot (dx\, i + dy\, j + dz\, k)$$

$$= \int_C 3xy\, dx - 5z\, dy + 10x\, dz$$

$$= \int_1^2 3(t^2+1)(2t^2)\, d(t^2+1) - 5(t^2)\, d(2t^2) + 10(t^2+1)\, d(t^2)$$

$$= \int_1^2 (12t^5 + 10t^4 + 12t^3 + 30t^2)\, dt = 303$$

Ex. (2): If $= 3xy\, i - y^2\, j$, evaluate $\int_C \mathbf{F} \cdot \mathbf{dr}$ where C is the curve in the xy plane, $y = 2x^2$, from *(0,0)* to *(1, 2)* ?

Solution: $\because$ *curve in the xy plane* $(z = 0) \implies \mathbf{r} = xi + yj$

$$\int_C \mathbf{F} \cdot \mathbf{dr} = \int_C (3xy\, i - y^2\, j) \cdot (dx\, i + dy\, j) = \int_C 3xy\, dx - y^2\, dy$$

$$\int_C \mathbf{F} \cdot \mathbf{dr} = \int_{x=0}^{1} 3x(2x^2)\, dx - (2x^2)^2\, d(2x^2)$$

$$= \int_0^1 (6x^3 - 16x^5)\, dx = -\frac{7}{6}$$

Ex. (3): If a force $\mathbf{F} = 2x^2y\, i + 3xy\, j$ displaces a particle in the *xy* plane from *(0, 0)* to *(1, 4)* along a curve $y = 4x^2$. Find the work done?

Solution: *work* done $= \int_C \mathbf{F} \cdot \mathbf{dr}$

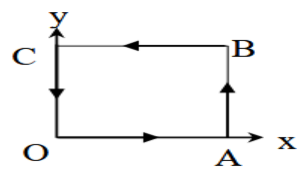
$$= \int_C (2x^2y\, i + 3xy\, j) \cdot (dx\, i + dy\, j)$$

$$= \int_C 2x^2y\, dx + 3xy\, dy = \int_{x=0}^{1} 2x^2(4x^2)\, dx + 3x(4x^2)\, d(4x^2)$$

$$= 104 \int_0^1 x^4\, dx = \frac{104}{5}$$

Ex. (4): Evaluate $\int_C \mathbf{F} \cdot \mathbf{dr}$ where $\mathbf{F} = x^2\, i + xy\, j$ and C is the boundary of the square in the plane $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$, $y = a$?

Solution: $\int_C \mathbf{F} \cdot \mathbf{dr} = \int_{OA} \mathbf{F} \cdot \mathbf{dr} + \int_{AB} \mathbf{F} \cdot \mathbf{dr} + \int_{BC} \mathbf{F} \cdot \mathbf{dr} + \int_{CO} \mathbf{F} \cdot \mathbf{dr}$



$\mathbf{r} = xi + yj \qquad \therefore \mathbf{dr} = dx\, i + dy\, j$

$\mathbf{F} \cdot \mathbf{dr} = (x^2\, i + xy\, j) \cdot (dx\, i + dy\, j)$

$\mathbf{F} \cdot \mathbf{dr} = x^2\, dx + xy\, dy \quad \ldots\ldots (1)$

On OA, $y = 0$, equation (1) become $\mathbf{F} \cdot \mathbf{dr} = x^2\, dx$

$$\therefore \int_{OA} \mathbf{F} \cdot \mathbf{dr} = \int_0^a x^2\, dx = \frac{a^3}{3} \quad \ldots\ldots (2)$$

On AB, $x = a$, $dx = 0$, equation (1) become $\mathbf{F} \cdot \mathbf{dr} = ay\, dy$

$$\therefore \int_{AB} \mathbf{F} \cdot \mathbf{dr} = \int_0^a ay\, dy = \frac{a^3}{2} \quad \ldots\ldots (3)$$

On BC, $y = a$, $dy = 0$, equation (1) become $\mathbf{F} \cdot \mathbf{dr} = x^2\, dx$

$$\therefore \int_{BC} \mathbf{F} \cdot \mathbf{dr} = \int_a^0 x^2\, dx = -\frac{a^3}{3} \quad \ldots\ldots (4)$$

On CO, $x = 0$, equation (1) become $\mathbf{F} \cdot d\mathbf{r} = 0$

$$\therefore \int_{CO} \mathbf{F} \cdot d\mathbf{r} = 0 \quad \ldots\ldots (5)$$

On adding equation (2), (3), (4) and (5), we get:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \frac{a^3}{3} + \frac{a^3}{2} - \frac{a^3}{3} + 0 = \frac{a^3}{2}$$

<u>Ex. (5):</u> A vector field is given by $\boldsymbol{F} = (2y+3)i + xz\,j + (yz - x)k$. Evaluate $\int \boldsymbol{F}.d\boldsymbol{r}$ along the path C is $x = 2t, y = t, z = t^3$ from $t = o\ to\ t = 1$?

<u>Solution:</u>

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \left((2y+3)\,i + xz\,j + (yz - x)\,k\right) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C (2y+3)dx + xz\,dy + (yz - x)\,dz$$

$$= \int_0^1 (2t+3)d(2t) + (2t)(t^3)\,d(t) + (t(t^3) - 2t)\,d(t^3)$$

$$= \int_0^1 (4t + 6 + 2t^4 + 3t^6 - 6t^3)dt = \frac{513}{70}$$

<u>Ex. (6):</u> (a) Show that **F** = $(2xy + z^3)i + x^2\,j + 3xz^2\,k$ is a conservative force field. (b) Find the scalar potential. (c) Find the work done in moving an object in this field from (1,-2,1) to (3,1,4)?

(a) $$\nabla \times \mathbf{F} = \begin{vmatrix} i & j & k \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix} = 0$$

Thus **F** is a conservative force field.

(b) $d\Phi = \boldsymbol{\nabla}\Phi\,.dr = \mathbf{F} \cdot d\mathbf{r}$

$$d\Phi = \left((2xy + z^3)i + x^2\,j + 3xz^2\,k\right) \cdot (dx\,i + dy\,j + dz\,k)$$

$$d\Phi = (2xy + z^3)dx + x^2\,dy + 3xz^2\,dz$$

$$\Phi = \int (2xy\,dx + x^2\,dy) + (z^3\,dx + 3xz^2\,dz)$$

$$\therefore \; scalar\ potential = \Phi = x^2y + xz^3 + constant.$$

(c) *work* done $= \int_{P_1}^{P_2} \mathbf{F} \cdot \mathbf{dr}$

$$= \int_{P_1}^{P_2} \left((2xy + z^3)i + x^2\,j + 3xz^2\,k\right) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_{P_1}^{P_2} (2xy + z^3)dx + x^2\,dy + 3xz^2\,dz$$

$$= \int_{P_1}^{P_2} (2xy\,dx + x^2\,dy) + (z^3\,dx + 3xz^2\,dz) = [x^2y + xz^3]_{P_1}^{P_2}$$

$$= [x^2y + xz^3]_{(1,-2,1)}^{(3,1,4)} = 202$$

Ex. (7): (a) Show that $F = 2x(y^2 + z^3)i + 2x^2y\,j + 3x^2z^2\,k$ is a conservative force field.

(b) Find the scalar potential. (c) Find the work done in moving an object in this field from $(-1,2,1)\ to\ (2,3,4)$?

Solution:

(a) $\nabla \times \mathbf{F} = \begin{vmatrix} i & j & k \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 2x(y^2 + z^3) & 2x^2y & 3x^2z^2 \end{vmatrix} = 0$

Thus **F** is a conservative force field.

(b) $\mathrm{d}\Phi = \nabla\Phi \,.\, dr = \mathbf{F} \cdot \mathbf{dr}$

$$\mathrm{d}\Phi = (2x(y^2 + z^3)i + 2x^2y\,j + 3x^2z^2\,k) \cdot (dx\,i + dy\,j + dz\,k)$$

$$\mathrm{d}\Phi = 2x(y^2 + z^3)\,dx + 2x^2y\,dy + 3x^2z^2\,dz$$

$$\Phi = \int (2xy^2\,dx + 2x^2y\,dy) + (2xz^3\,dx + 3x^2z^2\,dz)$$

$$\therefore \; scalar\ potential = \Phi = x^2y^2 + x^2z^3 + constant.$$

(c) *work* done $= \int_{P_1}^{P_2} \mathbf{F}\cdot d\mathbf{r} = \int_{P_1}^{P_2} d\Phi = [\Phi]_{P_1}^{P_2}$

$$= [x^2y^2 + x^2z^3]_{P_1}^{P_2}$$

$$= [x^2y^2 + x^2z^3]_{(-1,2,1)}^{(2,3,4)} = 287$$

<u>Ex. (8):</u> Evaluate $\iint (A.\hat{n})ds, where\ A = 18z\, i - 12\, j + 3y\, k$ and S is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant?

<u>*Solution:*</u>

$$\nabla(2x+3y+6z) = \left(i\frac{\partial\emptyset}{\partial x} + j\frac{\partial\emptyset}{\partial y} + k\frac{\partial\emptyset}{\partial z}\right)(2x+3y+6z) = 2i+3j+6k$$

$$\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{2i+3j+6k}{|2i+3j+6k|} = \frac{2}{7}i + \frac{3}{7}j + \frac{6}{7}k$$

$$\hat{n}\,.\,\hat{k} = \left(\frac{2}{7}i + \frac{3}{7}j + \frac{6}{7}k\right)\cdot\hat{k} = \frac{6}{7}$$

$$ds = \frac{dx\ dy}{(\hat{n}\,.\,\hat{k})} = \frac{dx\ dy}{\frac{6}{7}} = \frac{7}{6}dx\ dy$$

$$\therefore \iint_S (\mathbf{A}\cdot\hat{n})\, ds$$

$$= \iint_S \left((18z\, i - 12\, j + 3y\, k)\cdot\left(\frac{2}{7}i + \frac{3}{7}j + \frac{6}{7}k\right)\right)\frac{7}{6}dx\, dy$$

$$= \iint_S (6z - 6 + 3y)\, dx\ dy$$

From $2x + 3y + 6z = 12$, we get: $6z = 12 - 2x - 3y$

$$\therefore \iint_S (\mathbf{A}\cdot\hat{n})\, ds = \int_0^6 \int_0^{\frac{12-2x}{3}} (6-2x)\, dx\, dy$$

$$= \int_0^6 (6-2x)\, dx \int_0^{\frac{12-2x}{3}} dy = \int_0^6 (6-2x)\, dx \left[\frac{12-2x}{3}\right]$$

$$= \frac{1}{3}\int_0^6 (4x^2 - 36x + 72)\, dx = \frac{1}{3}\left[\frac{4x^3}{3} - 18x^2 + 72x\right]_0^6 = 24$$

Ex. (9): Evaluate $\iint (A.\hat{n})ds, where\ A = z\,i + x\,j - 3y^2 z\,k$ and S is that part of the plane $x^2 + y^2 = 16$ included in the first octant between $z = 0\ and\ z = 5$?

Solution:

$$\nabla(x^2+y^2) = \left(i\frac{\partial\emptyset}{\partial x} + j\frac{\partial\emptyset}{\partial y} + k\frac{\partial\emptyset}{\partial z}\right)(x^2+y^2) = 2xi + 2yj$$

$$\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{2xi+2yj}{|2xi+2yj|} = \frac{xi+yj}{4}$$

$$\hat{n}.\hat{j} = \left(\frac{xi+yj}{4}\right)\cdot\hat{j} = \frac{y}{4}$$

$$ds = \frac{dx\ dz}{(\hat{n}.\hat{j})} = \frac{dx\ dz}{\frac{y}{4}} = \frac{4}{y}dx\ dz$$

$$\therefore \iint_S (\mathbf{A}\cdot\hat{n})\,ds = \iint_S \left((z\,i + x\,j - 3y^2 z\,k)\cdot\left(\frac{xi+yj}{4}\right)\right)\frac{4}{y}dx\ dz$$

$$= \iint_S \left(\frac{xz+xy}{y}\right)dx\ dz$$

From $x^2+y^2=16$, we get: $y = \sqrt{16-x^2}$

$$\therefore \iint_S (\mathbf{A}\cdot\hat{n})\,ds = \int_0^5\int_0^4 \left(\frac{xz}{\sqrt{16-x^2}} + x\right)dx\,dz$$

$$= \int_0^5 \left[z\left(-\sqrt{16-x^2}\right) + \frac{x^2}{2}\right]_0^4 dz = \int_0^5 (4z+8)dz = 90$$

Ex. (10): If $\boldsymbol{F} = 2xz\, i - x\, j + y^2\, k$. Evaluate $\iiint \boldsymbol{F} \,.\, dV$, where V is the region bounded by the surface $x = 0, x = 2, y = 0, y = 6, z = x^2$ and $z = 4$?

Solution:

$$\iiint_V \mathbf{F}\, dV = \int_{x=0}^{2} \int_{y=0}^{6} \int_{z=x^2}^{4} (2xz\, i - x\, j + y^2\, k)\, dz\, dy\, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{6} [xz^2\, i - xz\, j + y^2 z\, k]_{x^2}^{4}\, dy\, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{6} (16x\, i - 4x\, j + 4y^2\, k) - (x^5\, i - x^3\, j + x^2 y^2\, k)\, dy\, dx$$

$$= \int_{x=0}^{2} \left[\left(16xy\, i - 4xy\, j + \frac{4}{3}y^3\, k\right) - \left(x^5 y\, i - x^3 y\, j + \frac{x^2 y^3}{3} k\right)\right]_0^6 dx$$

$$= \int_{x=0}^{2} (96x\, i - 24x\, j + 288\, k) - (6x^5\, i - 6x^3\, j + 72x^2\, k)\, dx$$

$$= \int_{x=0}^{2} (96x - 6x^5)\, i - (24x + 6x^3)\, j + (288 - 72x^2)\, k)\, dx$$

$$= \left[\left(\frac{96}{2}x^2 - x^6\right) i - \left(12x^2 + \frac{6}{4}x^4\right) j + (288x - 24x^3)\, k\right)\Big]_0^2$$

$$\therefore \iiint_V \mathbf{F}\, dV = (128)\, i - (24)\, j + (384)\, k$$

Ex. (11): If $\boldsymbol{F} = 2z\, i - x\, j + y\, k$. Evaluate $\iiint \boldsymbol{F} \,.\, dV$, where V is the region bounded by the surface $x = 0, x = 2, y = 0, y = 4, z = x^2$ and $z = 2$?

Solution:

$$\iiint_V \mathbf{F}\, dV = \int_{x=0}^{2} \int_{y=0}^{4} \int_{z=x^2}^{2} (2z\, i - x\, j + y\, k)\, dz\, dy\, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{4} [z^2\, i - xz\, j + yz\, k]_{x^2}^{2}\, dy\, dx$$

$$= \int_{x=0}^{2} \int_{y=0}^{4} (4\, i - 2x\, j + 2y\, k) - (x^4\, i - x^3\, j + x^2 y\, k)\, dy\, dx$$

$$= \int_{x=0}^{2} \left[(4y\, i - 2xy\, j + y^2\, k) - \left(x^4 y\, i - x^3 y\, j + \frac{x^2 y^2}{2} k\right)\right]_0^4 dx$$

$$= \int_{x=0}^{2} (16\, i - 8x\, j + 16\, k) - (4x^4\, i - 4x^3\, j + 8x^2\, k)\, dx$$

$$= \int_{x=0}^{2} (16 - 4x^4)\, i - (8x - 4x^3)\, j + (16 - 8x^2)\, k)\, dx$$

$$= \left[\left(16x - \frac{4}{5}x^5\right) i - (4x^2 - x^4)\, j + (16x - \frac{8}{3}x^3)\, k\right)\Big]_0^2$$

$$\therefore \iiint_V \mathbf{F}\, dV = \frac{32}{5} i + \frac{32}{3} j = \frac{32}{15}(3i + 5k)$$

<u>Ex. (12):</u> Find the volume of the region common to the intersecting cylinders $x^2 + y^2 = a^2\ and\ x^2 + z^2 = a^2$?

<u>Solution:</u>

$$\iiint_V \mathbf{F}\, dV = 8 \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \int_{z=0}^{\sqrt{a^2-x^2}} dz\, dy\, dx$$

$$= 8 \int_{x=0}^{a} \int_{y=0}^{\sqrt{a^2-x^2}} \sqrt{a^2 - x^2}\, dy\, dx = 8 \int_{x=0}^{a} (a^2 - x^2)\, dx$$

$$= 8 \left[a^2 x - \frac{x^3}{3}\right]_0^a = 8 \left(a^3 - \frac{a^3}{3}\right) = \frac{16\, a^3}{3}$$

Question 1:

If $\mathbf{A} = (3x^2+6y)\mathbf{i} - 14yz\,\mathbf{j} + 20xz^2\,\mathbf{k}$, evaluate $\int_C \mathbf{A}\cdot d\mathbf{r}$ from (0,0,0) to (1,1,1) along the following paths C:
(a) $x = t,\ y = t^2,\ z = t^3$.
(b) the straight lines from (0,0,0) to (1,0,0), then to (1,1,0), and then to (1,1,1).
(c) the straight line joining (0,0,0) and (1,1,1).

$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_C \left[(3x^2+6y)\mathbf{i} - 14yz\,\mathbf{j} + 20xz^2\mathbf{k}\right]\cdot(dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k})$$
$$= \int_C (3x^2+6y)\,dx - 14yz\,dy + 20xz^2\,dz$$

(a) If $x=t,\ y=t^2,\ z=t^3$, points (0,0,0) and (1,1,1) correspond to $t=0$ and $t=1$ respectively. Then

$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_{t=0}^{1} (3t^2+6t^2)\,dt - 14(t^2)(t^3)\,d(t^2) + 20(t)(t^3)^2\,d(t^3)$$
$$= \int_{t=0}^{1} 9t^2\,dt - 28t^6\,dt + 60t^9\,dt$$
$$= \int_{t=0}^{1} (9t^2-28t^6+60t^9)\,dt = 3t^3 - 4t^7 + 6t^{10}\Big|_0^1 = 5$$

Another Method.

Along C, $\mathbf{A} = 9t^2\mathbf{i} - 14t^5\mathbf{j} + 20t^7\mathbf{k}$ and $\mathbf{r} = x\mathbf{i}+y\mathbf{j}+z\mathbf{k} = t\mathbf{i}+t^2\mathbf{j}+t^3\mathbf{k}$ and $d\mathbf{r} = (\mathbf{i}+2t\mathbf{j}+3t^2\mathbf{k})\,dt$.

Then
$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_{t=0}^{1} (9t^2\,\mathbf{i} - 14t^5\,\mathbf{j} + 20t^7\,\mathbf{k})\cdot(\mathbf{i} + 2t\,\mathbf{j} + 3t^2\,\mathbf{k})\,dt$$
$$= \int_0^1 (9t^2 - 28t^6 + 60t^9)\,dt = 5$$

(b) Along the straight line from (0,0,0) to (1,0,0) $y=0,\ z=0,\ dy=0,\ dz=0$ while x varies from 0 to 1. Then the integral over this part of the path is

$$\int_{x=0}^{1} \left(3x^2+6(0)\right)dx - 14(0)(0)(0) + 20x(0)^2(0) = \int_{x=0}^{1} 3x^2\,dx = x^3\Big|_0^1 = 1$$

Along the straight line from (1,0,0) to (1,1,0) $x=1,\ z=0,\ dx=0,\ dz=0$ while y varies from 0 to 1. Then the integral over this part of the path is

$$\int_{y=0}^{1} (3(1)^2+6y)\,0 - 14y(0)\,dy + 20(1)(0)^2\,0 = 0$$

Along the straight line from (1,1,0) to (1,1,1) $x=1,\ y=1,\ dx=0,\ dy=0$ while z varies from 0 to 1. Then the integral over this part of the path is

$$\int_{z=0}^{1} \left(3(1)^2+6(1)\right)0 - 14(1)\,z(0) + 20(1)\,z^2\,dz = \int_{z=0}^{1} 20z^2\,dz = \frac{20z^3}{3}\Big|_0^1 = \frac{20}{3}$$

Adding,
$$\int_C \mathbf{A}\cdot d\mathbf{r} = 1 + 0 + \frac{20}{3} = \frac{23}{3}$$

(c) The straight line joining (0,0,0) and (1,1,1) is given in parametric form by $x = t, y = t, z = t$. Then

$$\int_C \mathbf{A}\cdot d\mathbf{r} = \int_{t=0}^{1} (3t^2+6t)\,dt - 14(t)(t)\,dt + 20(t)(t)^2\,dt$$

$$= \int_{t=0}^{1} (3t^2+6t-14t^2+20t^3)\,dt = \int_{t=0}^{1} (6t-11t^2+20t^3)\,dt = \frac{13}{3}$$

Exercise:

1.

If $\mathbf{A} = (2y+3)\mathbf{i} + xz\,\mathbf{j} + (yz-x)\mathbf{k}$, evaluate $\int_C \mathbf{A}\cdot d\mathbf{r}$ along the following paths C:

(a) $x = 2t^2,\ y = t,\ z = t^3$ from $t = 0$ to $t = 1$,

(b) the straight lines from (0,0,0) to (0,0,1), then to (0,1,1), and then to (2,1,1),

(c) the straight line joining (0,0,0) and (2,1,1).

Ans. (a) 288/35 (b) 10 (c) 8

2.

If $\mathbf{F} = (5xy-6x^2)\mathbf{i} + (2y-4x)\mathbf{j}$, evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$ along the curve C in the xy plane, $y = x^3$ from the point (1,1) to (2,8). *Ans.* 35